

Selective Fault-Tolerant RISC-V Processor with Error-Correcting Register File and Adaptive Redundancy Control

Area of the Project: ASIC Design

ABSTRACT

Introduction: Modern embedded systems deployed in drones, satellites, medical devices, and industrial controllers operate in harsh environments where radiation and electromagnetic interference can induce transient faults in digital logic. Conventional processors optimize for performance, area, and power but offer minimal protection against such faults, making them unsuitable for safety-critical applications. Existing fault-tolerant solutions typically replicate the entire processor, introducing prohibitive area, power, and complexity overhead incompatible with resource-constrained edge devices.

Problem Statement: There is a need for a lightweight, energy-efficient, and reliable processor that improves fault resilience without the excessive cost of full-system redundancy. Current approaches either leave systems unprotected or over-protect them, lacking a tunable trade-off between reliability and energy consumption.

Objectives: The primary objective is to design and verify a selective fault-tolerant RISC-V processor capable of maintaining correct operation during transient hardware faults. Specifically, the processor aims to: (i) execute the RV32E instruction set correctly; (ii) detect and correct memory errors in the register file using SEC-DED; (iii) mask computational faults in the execution stage using triple modular redundancy with majority voting; (iv) reduce power consumption by enabling redundancy only when required through clock gating; and (v) demonstrate the complete ASIC implementation flow using Cadence tools from RTL to GDSII.

Proposed Methodology: The design employs three complementary protection mechanisms: a (39,32) Hamming SEC-DED code on the 16-entry register file, TMR limited to the execution-stage ALU with a bit-wise majority voter, and single-bit parity on pipeline registers with automatic flush on detection. Hierarchical clock gating controls the redundant ALU slices, allowing dynamic power scaling with mission phase. The methodology follows a two-phase ASIC flow. Phase 1 (Semester 7) covers RTL design in SystemVerilog, functional verification in Cadence Xcelium with RISC-V compliance tests, fault-injection campaigns targeting the register file and ALU, logic synthesis in Cadence Genus with clock-gating insertion, and logical equivalence checking via Cadence Conformal LEC. Phase 2 (Semester 8) encompasses physical design in Cadence Innovus (floorplanning, placement, CTS, routing), parasitic extraction in Cadence Quantus, multi-corner static timing analysis in Cadence Tempus, power and IR-drop analysis in Cadence Voltus, and physical verification via DRC and LVS to produce a tapeout-ready GDSII. The expected outcome is a fully characterized processor with quantified area, timing, and power overheads alongside measured fault-coverage metrics.

References:

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